

2017/2018 Exam Paper Section B

Q2a) ii)
cont.

$$\pi_k = \frac{p_1}{p_2} \pi_{k-1} = p \pi_{k-1}$$

so (i) holds for k

$$\Rightarrow \pi_k = p^k \pi_0$$

We know $\sum_{k=0}^r \pi_k = 1$

$$\Rightarrow \sum_{k=0}^r p^k \pi_0 = 1$$

$$\Rightarrow \pi_0 = \frac{1}{\sum_{k=0}^r p^k}$$

$$\Rightarrow \pi_0 = \frac{1}{\frac{1-p^{r+1}}{1-p}} = \frac{1-p}{1-p^{r+1}}, \quad p \neq 1$$

$$\pi_0 = \frac{1}{r+1}, \quad p = 1$$

Q2b) $\alpha_k = \alpha$, $\beta_k = k\beta$, $\alpha > 0$, $\beta > 0$

Show π_k , $k=0, 1, 2, \dots, r$ are given by

$$\pi_k = \frac{p^k}{k! \cdot \frac{1 + p + \frac{p^2}{2!} + \dots + \frac{p^r}{r!}}}{\text{where } p = \frac{\alpha}{\beta}}$$

$$\pi_k = \frac{p^k}{k!} \pi_0, \quad \pi_k = \frac{p^k}{k!} e^{-p}$$

Since $\sum_{k=0}^r \pi_k = 1$ we get $\pi_0 \sum_{k=0}^r \frac{p^k}{k!} = 1$

$$\pi_0 = \frac{1}{\sum_{k=0}^r \frac{p^k}{k!}}$$

$$\Rightarrow \pi_k = \frac{p^k}{k!} e^{-p} \cdot \frac{1}{\sum_{i=0}^r \frac{p^i}{i!}} \quad k=0, 1, \dots, r.$$

←

2017/2018 Exam Paper Section B

Q2a) ii)
cont.

$$\pi_k = p_1 \pi_{k-1} = p_2 \pi_{k-1}$$

so (i) holds for k

$$\Rightarrow \pi_k = p^k \pi_0$$

We know $\sum_{k=0}^r \pi_k = 1$

$$\Rightarrow \sum_{k=0}^r p^k \pi_0 = 1$$

$$\Rightarrow \pi_0 = \frac{1}{\sum_{k=0}^r p^k}$$

$$\Rightarrow \pi_0 = \frac{1}{\frac{1-p^{r+1}}{1-p}} = \frac{1-p}{1-p^{r+1}}, \quad p \neq 1$$

$$\pi_0 = \frac{1}{r+1}, \quad p = 1$$

Q2b) $\alpha_k = \alpha$, $\beta_k = k\beta$, $\alpha > 0$, $\beta > 0$

Show π_k , $k=0, 1, 2, \dots, r$ are given by

$$\pi_k = \frac{p^k}{k!} \frac{1}{1 + p + \frac{p^2}{2!} + \dots + \frac{p^r}{r!}} \quad \text{where } p = \frac{\alpha}{\beta}$$

$$\pi_k = \frac{p^k}{k!} \pi_0, \quad \pi_k = \frac{p^k}{k!} e^{-p}$$

Since $\sum_{k=0}^r \pi_k = 1$ we get $\pi_0 \sum_{k=0}^r \frac{p^k}{k!} = 1$

$$\pi_0 = \frac{1}{\sum_{k=0}^r \frac{p^k}{k!}}$$

$$\Rightarrow \pi_k = \frac{p^k}{k!} e^{-p} \frac{1}{\sum_{i=0}^r \frac{p^i}{i!}} \quad k=0, 1, \dots, r.$$

←

Q2 b)
cont.

$$\text{Then } \pi_k = \frac{\rho^k}{k!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \dots + \frac{\rho^r}{r!}}$$

Q2 c)

phone exchange with 3 operators
30 people call the exchange per hour
each call is serviced in 4 minutes
arrivals = α , services = β

$$\alpha = \frac{30}{60} \text{ min}^{-1} = \frac{1}{2} \text{ min}^{-1}$$

$$\beta = \frac{1}{4} \text{ min}^{-1}$$

$$\rho = \frac{\alpha}{\beta} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

$$r = 3$$

$$\pi_3 = \frac{\rho^3}{3!} \frac{1}{1 + \rho + \frac{\rho^2}{2!} + \frac{\rho^3}{3!}}$$

$$\pi_3 = \frac{4}{3} \left(\frac{3}{19} \right)$$

$$\pi_3 = \frac{4}{19}$$

2017/2018 Exam Paper Section B

Q3 a) r is not the growth rate. Provided the $P < r$ then $R(P) < 1$ and population declines.

Q3 b) $P_{n+1} = R(P_n)P_n$,

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}$$

Fixed points where $R(P) = 1$

i.e. $P^2 - rK^2P + K^2 = 0$

$P_{n+1} = R(P_n)P_n$

$$R(P) = \frac{rP}{1 + \left(\frac{P}{K}\right)^2}$$

if $0 < P \ll K$ then $R(P) = rP$

$\Rightarrow P_{n+1} = rP_n^2$

$P^* = 0$ is a trivial fixed point

Let $R(P) = 1 \Rightarrow \frac{rP}{1 + \left(\frac{P}{K}\right)^2} = 1$

$$rP = 1 + \left(\frac{P}{K}\right)^2$$

$$P^2 - K^2rP + K^2 = 0$$

two roots = P^* & P_{crit} $P^* > P_{crit}$

$(P - P^*)(P - P_{crit}) = 0$

$R(P) < 1 \Rightarrow \frac{rP}{1 + \left(\frac{P}{K}\right)^2} < 1$

$$\Rightarrow \frac{P^2}{K^2} - rP + 1 > 0$$

$(P - P^*)(P - P_{crit}) > 0$

$P < P_{crit} \Rightarrow R(P) < 1$

$$P^* = \frac{1}{2}rK^2 \left(1 + \sqrt{1 - \left(\frac{2}{rk}\right)^2} \right)$$

$$P_{crit} = \frac{1}{2}rK^2 \left(1 - \sqrt{1 - \left(\frac{2}{rk}\right)^2} \right)$$

Q3 c) i)

$$r = 2 \quad k = 10 \quad P_0 = 1 \quad R(P) > 1$$

$$P^* = \frac{1}{2} (2)(10)^2 \left(1 + \sqrt{1 - \left(\frac{2}{(2)(10)} \right)^2} \right) = 199.499$$

$$P_{crit} = \frac{1}{2} (2)(10)^2 \left(1 - \sqrt{1 - \left(\frac{2}{(2)(10)} \right)^2} \right) = 0.501$$

$$P_0 = 1$$

$$P > P_{crit}$$

$$1 > 0.501$$

$0.501 < 1 < 199.499 \Rightarrow$ population increases and is bounded above by P^*

Q3 c) ii)

$$P_0 = 50 \quad R(P) > 1$$

$$P > P_{crit}$$

$$0.501 < 50 < 199.499$$

$\Rightarrow$ population increases & is bounded above by P^*

Q3 c) iii)

$$P_0 = 100 \quad R(P) > 1$$

$$P > P_{crit}$$

$$0.501 < 100 < 199.499$$

$\Rightarrow$ population increases & is bounded above by P^*

